

Qatar-1b

R-1946

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_131022 · created 2026-08-06T15:05:23+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456587.68896 +/- 0.00015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.191 +/- 0.022
Transit depth (Rp/Rs)^2	3.66 +/- 0.86 [%]
Orbital Inclination (inc)	82.17 +/- 0.96
Airmass coefficient 1 (a1)	1.27 +/- 0.022
Airmass coefficient 2 (a2)	-0.154 +/- 0.018
Scatter in the residuals of the lightcurve fit is	1.77 %
Best Comparison Star	None
Optimal Aperture	2.79
Optimal Annulus	5.8
Transit Duration (day)	0.06 +/- 0.0086

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.191 $\pm$ 0.022 vs 0.1463 (deviation 2.03 σ)
Duration fit vs published	0.06 d vs 0.0692 d (deviation 1.07 σ)
Mid-time O-C	-0.4 min
Depth SNR	8.7
Residual scatter	1.77 %

Light curve

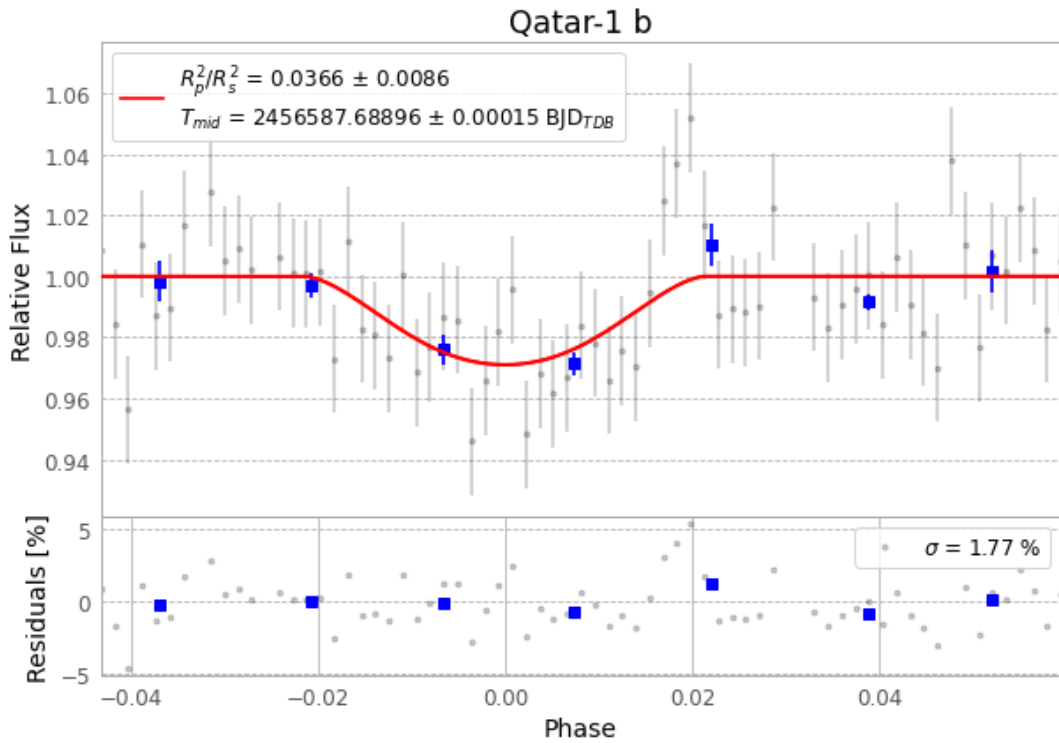


Figure 1 — FinalLightCurve_Qatar-1 b_2013-10-21.png (SHA-256 e32e9b1aad66811b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [378, 289], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b53ca0af2477cc5b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

71 FITS frames (47 MB), Qatar-1131022030012.FITS → Qatar-1131022063012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-131022.log (03ba7c1882a7...), FinalLightCurve_Qatar-1 b_2013-10-21.png (e32e9b1aad66...), FinalLightCurve_Qatar-1 b_2013-10-21.csv (91efab026309...)

Compute resources

Wall time 148.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 15:05Z.